An interdisciplinary review and concept by Romul Protocol

MACHINE-CENTRIC MARKETING

Where a new marketing logic fits in the age of AI agents and machine buyers, and why machine-verifiable honesty is its foundation

A concept paper on marketing in which the one who receives the message and makes the choice is no longer only a person but also a machine: an AI agent or another automated system. The paper brings together how several fields see the same process (marketing, the economics of information, research on AI agents, and Ethereum architecture) and proposes two linked working concepts: machine-centric marketing as the task, and machine-verifiable honesty as the property of the environment that gives machine trust something to stand on.

Author

Human and Machine

Current as of

State of terminology and the market
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Abstract and scope

This paper looks at a process that is only starting to be studied: a share of buyers in the market are becoming machines, that is, AI agents acting on behalf of people and organizations. Today several fields describe this process separately. Marketing and law talk about the machine customer and the algorithmic consumer. Economics of information talks about asymmetry, signals, verifiable disclosure and credible commitments. Research on AI agents runs experiments with agent buyers. Cryptography and Ethereum engineering work on trust minimization, verifiable credentials and agent payment protocols. Each field sees its own part of the picture and rarely cites the others.

In genre, this is **an interdisciplinary review with the author's own concept**. The first goal is to gather in one place, with sources, what each of these fields already knows about the machine buyer and about trusting it. The second is to offer a framework that ties them together, built on two concepts: **machine-centric marketing** and **machine-verifiable honesty**. The paper does not claim to be fundamental research and reports no experiments of its own. Its value lies in bringing different fields to bear on one process and in the ideas that follow from doing so.

> **How to read this paper.** The review part relies on sources, cited in square brackets. The author's own ideas and proposals are set apart in blocks marked "Author's note." On the empirical side, the paper relies on other people's research.

1 Main conclusion

> **"Machine-centric marketing"** is a strong term for a distinct logic within marketing as a whole, if we take it to mean marketing optimized not merely with the help of machines but **for the machine that takes part in choosing and buying**. Above all, this means AI agents that research the market, compare offers, filter options and, in some scenarios, complete the purchase on behalf of a person or an organization.

> **The key shift in perspective:** traditional marketing mostly optimizes communication with a person. Machine-centric marketing has to optimize for something else: getting the product into the machine's "consideration set," having it interpreted correctly, compared fairly and, in the end, chosen by the machine.

> **Three levels worth keeping apart:** Agentic Commerce as the broader model of trade; Machine Customer as a new type of buyer, a new participant on the demand side; Machine-Centric Marketing as the proposed name for the marketing discipline that forms around this buyer.

Machine trust needs a foundation: a machine does not take claims on faith, it checks them. So the central question of machine-centric marketing is not "how do we persuade?" but "how do we make checking cheap?" We call an environment where checking a claim costs almost nothing, and deception is either technically impossible or immediately visible, **machine-verifiable honesty** (Section 8). Machine Trust rests on it, and it gives Machine Persuasion a precise form. This is a normative claim. Today's agents can still be swayed by presentation (Section 2.3), and that is exactly why they need an environment where verification is cheaper than trust.

2 Review: four lines of research on one process

What follows is a short summary of what each of the four fields already knows about the machine buyer and about trusting it. This is not a systematic review. It names the key works that the rest of the paper builds on.

2.1. Marketing and law: the machine buyer

One of the first academic formulations came from legal scholars. Gal and Elkin-Koren (2017) described "algorithmic consumers," digital agents that predict needs, make purchase decisions, negotiate and complete the transaction for a person, and they analyzed what this means for competition and regulation. [12] Dawar and Bendle (Harvard Business Review, 2018) carried the idea into marketing: as assistants satisfy consumers, trust in the assistant starts to replace trust in the brand, and marketing becomes a battle for the AI assistant's attention. [13] Gartner has been developing the idea of the machine customer since 2015. In the book by Scheibenreif and Raskino (2023), a machine customer is defined as a non-human economic actor that obtains goods or services in exchange for payment, and the authors describe three phases of its evolution: "bound," "adaptable" and "autonomous" customers. [14]

Closest to the practice of machine visibility is the research on Generative Engine Optimization (Aggarwal et al., KDD 2024): adding citations, quotations and statistics makes a source much more visible in a generative system's answers, while classic keyword stuffing brings little to no improvement and, on the deployed Perplexity engine, actually made things worse. [15] In other words, signs of verifiability already attract the machine reader today. Academic work on the machine buyer as the audience of marketing is ongoing (Accornero, 2026 [4]), and in 2026 the first call for a separate discipline appeared in a peer-reviewed marketing journal. Sarstedt, Adler and Imschloss (Journal of Marketing Analytics) introduce the "AI customer" as a distinct market actor with its own decision logic and argue that marketing needs a new subdiscipline, "machine marketing," devoted to understanding, predicting and shaping the behavior of AI-based decision-makers. [52] Their research note sets out the agenda; this paper takes up one question inside it: the role of verification and what it costs.

2.2. The economics of information and trust

The problem that machine-verifiable honesty addresses in this paper was formalized in economics long ago. Akerlof (1970) showed how information asymmetry about quality destroys a market. [17] Spence (1973) explained why a signal carries information only when it costs the honest seller less than the dishonest one. [18] Grossman (1981) and Milgrom (1981) proved the "unraveling" result: if claims can be verified and the buyer is skeptical, silence reads as a bad sign, and sellers are better off disclosing everything. [19][20] At the other end is "cheap talk" as described by Crawford and Sobel (1982), where

messages that cannot be verified convey only limited information [21]; the modern theory of Bayesian persuasion (Kamenica and Gentzkow, 2011) describes how a sender designs the information environment of the receiver. [22]

The second classic line is about commitments. Williamson (1983) showed how "hostages," losses accepted voluntarily in case of breach, make promises credible. [23] Klemperer (1995) laid out the role of switching costs in competition. [24] In relationship marketing, the central work is the commitment-trust theory of Morgan and Hunt (1994), in which commitment and trust are the key mediators of long-term relationships. [16] Finally, Shahidi, Rusak, Manning, Fradkin and Horton (NBER, 2025) analyze how AI agents lower the transaction costs of search, communication and contracting, and what new frictions appear along the way. [25]

2.3. How AI buyer agents behave: the evidence

Since 2025, controlled experiments with agent buyers have started to appear. The ACES study (Columbia University; ACM Web Conference 2026) found that agents' choices cluster on a few products, their preferences shift between model versions, and they show strong position biases. They penalize sponsored tags and reward platform endorsements, and simple description tweaks made by a seller-side agent can drive significant gains in market share. [26] Magentic Marketplace (Microsoft Research, 2025) found a strong "first-proposal bias" and tested three manipulation tactics against agents: psychological persuasion, fake credentials and prompt injection. Prompt injection worked, and even basic persuasion proved effective. [27] Cherep, Maes and Singh (2025) showed that LLM agents are more sensitive than people to "nudges" such as defaults, suggestions and highlighted information. [28] Experiments in credence goods markets show that making claims verifiable does not by itself guarantee honest behavior or higher welfare among LLM agents. [29]

At this point the review leads to a clear conclusion: today's agents can be swayed by presentation, and they can be manipulated. This matters for everything that follows (Sections 8.4 and 12).

2.4. Cryptography, Ethereum and the trust infrastructure of agentic commerce

The idea of trust minimization goes back to Szabo: a trusted third party built into a protocol is a security hole installed in advance. [8] The first working system built on this idea was Bitcoin. From its first lines, Nakamoto describes it as an electronic payment system based on cryptographic proof instead of trust. [51] In a sense, the whole line of thought this paper follows starts here. Buterin formulated the principle of credible neutrality: a mechanism is neutral if you can see that from its design, and its execution is open and publicly verifiable. [7] Douceur (2002) showed that without a logically centralized authority, Sybil attacks are always possible except under extreme and unrealistic assumptions. [9] Proof of solvency (Dagher et al., 2015) showed how to prove solvency without revealing all the data. [37] Simplicity as a condition of security goes back to Saltzer and Schroeder (1975) and their principle of "economy of mechanism": keep the design as simple and small as possible. [39]

In 2025 and 2026 this line turned into industry standards for agentic commerce. W3C Verifiable Credentials 2.0 (2025) defines cryptographically secured, machine-verifiable credentials. [33] Google AP2 (2025) builds trust in agent payments on "mandates," tamper-proof, cryptographically signed digital contracts that serve as verifiable proof of the user's instructions [34]; Mastercard Verifiable Intent (2026) creates a tamper-resistant record of what the user authorized. [35] Stripe and OpenAI (ACP) and Coinbase (x402) standardize the agent's payment itself [38]; they have been joined by the Machine Payments

Protocol (Stripe and Tempo) for paying for APIs and other web resources, and by Visa's Trusted Agent Protocol, which helps merchants recognize trusted agents. [42][49][50] On Ethereum, the ERC-8004 "Trustless Agents" standard introduces registries for agent identity, reputation and validation. [36]

> **Where the gap is:** all of these standards verify the *buyer's* side and the transaction itself, that is, who authorized what and whether it was carried out. Verifying what the *seller* claims about its own state, rules and commitments is far less standardized. This is the gap that machine-verifiable honesty is aimed at.

In September 2026, the world's largest asset manager also recorded the convergence of these lines. BlackRock's report "The Machine-Native Economy" describes AI as machine-native intelligence and digital assets as machine-native money. In the report's words, tokenized assets are designed for machine-verifiable transfer and settlement, while KYC, AML and KYA (know-your-agent) checks happen off-chain, and only the verified results are passed on-chain. [42] This is industry material that itself states it is not research or investment advice, and its focus is once again on payments and settlement. The authors admit that agent payments are still at an early stage, and independent data confirms it: TRM Labs estimates that of roughly $52.7 million processed through x402 since May 2025, AI agents account for only 0.6 to 7.5% of likely commerce. [43]

2.5. The closest work

The works closest to the ideas in this paper appeared in 2025 and 2026, and all of them are preprints. Ventirozos and Shardlow (2026) show that when the buyer is an agent, what's scarce is no longer ranking but trustworthy information, and the competitive question shifts from "who ranks first" to "who can substantiate their claims most cheaply and credibly"; in their model, verification is a paid microtransaction. [30] Xie (2026) builds a game-theoretic model of buyer and seller agents with verification and shows that verification helps only when it covers the disputed proposition, and that it can be privately underprovided or, in other cases, overprovided; in the model, verified execution does not prove true value. [31] Madugula, Esteva de la Rosa and Shankar (2026) build verifiable execution into a model of loyalty under agentic commerce. [32]

The same idea shows up independently in practical business writing. Katja Forbes's book "Machine Customers: The Evolution Has Begun" (2025) starts from the premise that machine customers don't respond to emotional appeals or brand storytelling; they evaluate clear information, consistent performance and verifiable claims. [44] Accornero, author of the demand-side work on the machine customer [4], puts it even more bluntly in the announcement of his book "The Algorithmic Shopper" (scheduled for April 2027): the only way to appeal to an AI agent is to be verifiably better, that is, to have a better product, a more transparent supply chain and cleaner, more accessible data. [45] Finally, McKinsey's analysis of agentic commerce concludes that loyalty becomes less about sentiment and more about policy [5], which lines up directly with this paper's claim that verifiable commitments replace loyalty (Section 9.3).

Author's note: where this paper stands. This paper came about independently, out of the author's hands-on practice as a programmer and marketer, and only later was it compared with the literature. We take the fact that different fields reached similar conclusions by different paths as an argument that the process described here is real. Our approach differs in that it treats the cost of verification not as a service and not as an equilibrium choice between parties, but as **a design parameter of the environment**, one that can be brought close to zero for claims about state and rules, and as **a measurable marketing quantity**. The closest academic framing, machine marketing as proposed by Sarstedt and colleagues [52], argues that a new subdiscipline is needed; we offer an answer to what its foundation could be. The contribution is not the discovery of individual mechanisms, each of which is known, but bringing them into one framework where marketing, the economics of trust and the architecture of verifiable systems describe the same shift.

3 What already exists in international terminology

Judging by current sources from 2026, the terminology is taking shape but has not yet settled into a single standard. Some terms are catching on quickly, while a name specifically for "marketing aimed at the machine as the audience" is still open.

Term	Meaning	Status	Relation to this concept
Agentic Commerce	Buying and selling in which AI agents research, compare, negotiate and/or complete deals on behalf of people or businesses.	Already catching on noticeably.	The broad frame; covers more than marketing.
Machine Customer	A machine economic actor able to act as a buyer: to find, evaluate and purchase.	Still forming, but quickly gaining meaning.	Almost a direct description of marketing's new audience.
Agent-to-Agent Commerce	Interaction between a buyer's AI agent and a seller's AI agent.	Actively used in B2B and agentic commerce.	Describes the channel and architecture of interaction, not marketing itself.
Machine-Readable Commerce	Commercial information and interfaces structured so that machines can interpret them reliably.	An applied, growing term.	Describes the technical layer that machine-centric marketing needs.
Agentic Marketing	Marketing that uses AI agents and autonomous automation.	Meaning still broad and ambiguous.	Needs to be kept distinct from marketing aimed at AI agents.
GEO / Generative Engine Optimization	Optimizing the presence of a brand or content in generative search and answer systems.	A parallel practice.	Part of the visibility task, but not the whole system of machine-centric marketing.
Machine Marketing / AI Customers	A proposed marketing subdiscipline devoted to understanding, predicting and shaping the behavior of AI-based decision-makers ("AI customers").	Proposed in a peer-reviewed research note, 2026. [52]	The closest academic framing; machine-centric marketing adds verifiability as its foundation (Section 8).

Term	Meaning	Status	Relation to this concept
Verifiability / Trust Minimization / Credible Neutrality	Verifiable execution, minimal trust in intermediaries, mechanisms whose neutrality is visible from their design.	Established concepts in the design of decentralized mechanisms. [7][8]	The conceptual base of machine-verifiable honesty (Section 8).

What this adds up to: international practice already has enough terms to describe the new commercial environment, but "marketing for machine customers" in particular has no single accepted, established name; the first academic proposal, "machine marketing," appeared only in 2026. [52] The concepts of verifiability and trust minimization, by contrast, are well developed, but in mechanism design rather than in marketing theory.

4 Related concepts and where they end

4.1. Agentic Commerce: the overall frame

IBM defines agentic commerce as an approach to buying and selling in which AI agents act on behalf of consumers or businesses to research, negotiate and complete purchases, often without direct human intervention. [1] Deloitte describes the same shift as a journey: from assisted discovery and assisted shopping to agentic shopping, then autonomous shopping, and finally agent-to-agent commerce, as consumers hand over more and more purchasing authority to agents. [2][3]

The essential difference: Agentic Commerce answers the question "how will trade happen?", while machine-centric marketing answers "how do we get the machine to choose our product in this new trading environment?"

4.2. Machine Customer: a new type of buyer

The term Machine Customer is especially close to the proposed model; in industry analysis, Gartner has been developing it since 2015. [14] It captures a change on the demand side: the buying subject becomes a machine acting on behalf of a person or an organization. In a 2026 academic paper, Paul F. Accornero directly proposes treating the AI agent as a new kind of "buyer" in categories where the purchase decision and its execution are delegated to a machine. [4]

This lets us distinguish three roles:

Consumer	the person who experiences the value and forms the underlying need.
Shopper / Buyer	the one who actually searches, compares and chooses the offer.
Machine Customer / AI Agent	a machine participant that can perform the shopper or buyer role on behalf of a person or an organization.

4.3. Agent-to-Agent Commerce: the machine channel for deals

In B2B, the "agent-to-agent" scenario is described more and more explicitly: the buyer's agent and the seller's agent interact directly, explore options, agree on terms and can trigger transactions. Deloitte calls this a possible next stage of B2B commerce. [3]

This matters for marketing because the classic chain "brand → person → purchase" may turn into "brand → machine agent → purchase," or even "seller's agent ↔ buyer's agent → deal." The limiting case comes when the seller's side is represented by a public, executable rule instead of an agent with its own discretion. Its behavior is fully predictable, and the buying agent only needs to check the rule itself.

4.4. Machine-Readable Commerce: the technical layer

If a machine takes part in the choice, a good-looking product page isn't enough for it. It needs unambiguous entities, attributes, prices, availability, constraints, variants, policies, evidence and a clear path to action. This practical side is increasingly described as machine-readable commerce. [5] EY puts it more firmly: companies need commercial data and terms that systems can interpret without ambiguity, with products, prices, rules and promises clear and consistent enough for machines to trust. [6] According to BlackRock, standardized machine-readable representations of assets reduce the need for custom integrations and let agents run more complex chains of actions. [42]

So machine-centric marketing includes information architecture: being "visible" isn't enough for a product; a machine also has to be able to read, compare and verify it correctly. Readability, though, is not yet verifiability: a structured claim is still a claim until the machine can check it against a primary source. That next step is covered in Sections 8 and 9.

4.5. Agentic Marketing: not the same thing

Agentic Marketing may refer to marketing carried out with the help of AI agents: generating campaigns, automated optimization, autonomous process launches, personalization and other forms of automation. This field sees the machine as the one doing the marketing. Machine-centric marketing sees the machine as the audience and a participant in the choice.

> **The difference in one line:**
> Agentic Marketing = "marketing that a machine does."
> Machine-centric marketing = "marketing that a machine has to understand and accept."

4.6. GEO / Generative Engine Optimization: only one of the tools

Optimizing presence in generative search is close to the visibility task, but narrower in scope. GEO focuses on getting a brand or content to appear in a generative system's answers. Machine-centric marketing has to cover the whole path, from discovery to comparison, trust, choice and the deal.

5 A proposed conceptual architecture

What follows is a working model, not an industry standard. Its job is to fit the term logically into general marketing theory.

MARKETING	Creating, communicating and delivering value in the market
→ **Human-Centric Marketing**	Optimizing for a person's perception, motivation, trust and behavior
→ **Machine-Centric / Machine-Oriented Marketing**	Optimizing for how a machine agent perceives, evaluates and chooses a product
→ **Agentic Commerce**	The broader context in which such a machine choice can end in autonomous trade
→ **Machine Customer**	The specific type of machine participant on the demand side
→ **Machine-Verifiable Honesty**	The property of the environment that gives machine trust a basis: verification is cheaper than trust, and deception is impossible or immediately visible

Note: "Machine-Centric Marketing," "Machine-Oriented Marketing" and "Machine-Verifiable Honesty" are offered here as working names for the concept. Searching current English-language sources, we found no sign that any of them has already become a widely used international term with a stable, recognized definition. The closest existing term is "machine marketing." [52]

6 A working definition

Machine-Centric Marketing: the branch of marketing concerned with designing, presenting and distributing value so that AI agents and other machine systems can discover, correctly interpret, compare, evaluate and choose an offer, acting as an intermediary for, or the direct executor of, the purchase decision of a person or an organization.

The name. We chose Machine-Centric Marketing (MCM) as the main term because it best conveys the idea of marketing built around the machine participant in the choice. Machine-Oriented Marketing is acceptable as a synonym but sounds more technical. The foundation of the concept is called Machine-Verifiable Honesty (MVH). Both names are offered as working names, not as an existing standard (Section 5). The term is deliberately not tied to today's technology: it covers AI agents, recommender systems, procurement algorithms and other mechanisms of automated choice.

The key word in this definition is "marketing." This is not only about technically integrating a store with an agent's API; it's about the full set of marketing tasks: positioning, value proposition, proof, trust, differentiation, visibility, conversion and retention, keeping in mind that the machine does much of the cognitive work.

Of these tasks, one becomes decisive when the audience is a machine: trust. A person can be brought to trust through image, reputation and social proof. A machine acting on its owner's behalf shouldn't take anything on faith; its job is to check claims against sources. That's why the definition is paired with a second concept, which describes what the environment the seller operates in must be like: machine-verifiable honesty (Section 8).

7 What machine-centric marketing includes

Machine Visibility.	Getting the product into the pool from which an AI agent picks its options.
Machine Interpretability.	An unambiguous understanding of what the product is, its properties, price, terms and constraints.
Machine Comparability.	Presenting the product in a form that lets the agent compare it correctly with the alternatives.
Machine Trust.	Evidence, data provenance, guarantees, policies and other signals the machine can use to reduce uncertainty. At its limit, this is an environment of machine-verifiable honesty, where a claim is replaced by a readable fact (Sections 8 and 9).
Machine Persuasion.	Not "emotionally persuading a machine," but creating the kind of value signals and evidence that shape how a machine ranks and chooses. The strongest signal for a machine is one that costs almost nothing to verify (Section 8.4).
Agent Conversion.	Designing the path by which an agent's recommendation turns into action: adding to cart, checking out, requesting a quote or closing the deal.
Agent-to-Agent Selling.	Preparing the commercial offer and the interaction logic for the case where both the buyer and the seller are represented by agents.

8 Machine-verifiable honesty: the foundation of machine trust

8.1. The problem: claims about oneself

In any economic system, participants make claims about themselves: "we have this much inventory and liquidity," "we won't change the terms," "there's real demand behind our product." Traditionally, such claims are either taken on trust or checked expensively and after the fact. Trust is a cost and a vulnerability: wherever you have to trust, you can be deceived. The literature on protocol security puts it bluntly: a trusted third party built into a design is a security hole installed in advance, which will later have to be closed by other means. [8]

For a human buyer, brand and reputation partly remove the cost of trust. For a machine buyer, this path shouldn't work: an agent protecting its owner's interests has to rely on data, not on presentation. For such an agent, a claim that can't be verified is either noise or risk.

8.2. Definition and the asymmetry of costs

> **Machine-Verifiable Honesty:** the property of a system designed so that verifying claims costs almost nothing, while deception is either technically impossible or immediately visible.

Strong and weak forms. In the strong form, the facts themselves are verifiable: state and rules exist inside a verifiable environment, as with digital and financial assets on public blockchains. In the weak form, only the origin of claims about the outside world is verifiable: who signed them and when, as with verifiable credentials for a product or service. [33] The strong form serves as the reference point and is the main subject of Sections 8 and 9; the weak form describes what is achievable for most traditional commerce (Section 11.4).

The concept rests on ideas described in the literature (verifiability, trust minimization, credible neutrality) and, originally, on the principle of "proof instead of trust" that Bitcoin started from. [7][8][51] At its core is an asymmetry of costs:

> The system is built so that honesty becomes the path of least resistance. Not "deception is forbidden," but "deception costs more and makes less sense than honesty, because verifying is almost free and deceiving is nearly impossible or immediately noticed."

The emphasis moves from the technical possibility of verification to the economics of honesty: honesty here is not a participant's virtue and not a prohibition, but a property of an environment in which a rational participant is honest by default.

In terms of economic theory, the asymmetry of costs is an informal statement of several known results: verifiable disclosure makes silence suspicious and leads to full disclosure (Grossman, Milgrom [19][20]); a signal carries information when faking it costs the dishonest participant more (Spence [18]); a mechanism in which honest behavior pays better is called incentive-compatible in mechanism design. The seller's choice doesn't disappear; it moves to the design stage, where the seller decides whether to build on a verifiable foundation and which parameters to freeze. In essence, this is a credible commitment in Williamson's sense [23], made once, when the system is created. What machine-verifiable honesty adds to this line of work is not a new mechanism but a change of focus and the transfer of these results into marketing: once such a commitment is made, disclosure is no longer a separate decision in every deal and becomes a *state* of the environment, one in which nothing goes undisclosed.

8.3. Honesty as a property of the environment, not the participant

Ordinary honesty is a property of an agent: "he doesn't lie." Machine-verifiable honesty is a property of an environment: "lying here makes no sense, because everything can be checked, and checking is cheaper than trusting." Designing honest participants means relying on the most fragile foundation there is. Designing an environment where honesty is rational means relying on a stable one.

For marketing, this changes the object of the work: the subject is no longer a message about honesty, but a setup that makes such a message unnecessary.

8.4. Persuading a machine means lowering the cost of verification

Empirical studies (Section 2.3) show that today's AI agents *do* respond to presentation: position in the results, the wording of a description, fake credentials and instructions embedded in text noticeably affect their choices. [26][27][28] So the claim "an agent can't be swayed by presentation" is wrong as a description of how things are.

> **Author's note.** The thesis of this paper is normative: an agent designed as a verifier *should not* give in to what can't be verified, and an environment of machine-verifiable honesty *reduces the payoff* from presentation and manipulation. The weakness of today's agents doesn't refute the thesis; it motivates it: the easier an agent is to manipulate, the more valuable an environment where verification is cheaper than trust becomes for it. A person goes from "persuade" to "believe." An agent that is supposed to protect its owner's interests goes from "show" to "check."

Hence the practical rule of machine-centric marketing: don't amplify the claim, make it cheaper to verify. Other things being equal, an offer whose properties an agent can check with a single request to a primary source should win over an offer that requires trust, as agents become more cautious and get verification tools. This is close to the conclusion of Ventirozos and Shardlow that the competitive question is shifting toward who "can substantiate their claims most cheaply and credibly." [30] The difference between looking verifiable and being verifiable (Section 11.6) is central to the ethics of machine-centric marketing (Section 12).

8.5. Two levels of honesty

The state level: "what is there now, is visible." Any claim about the current situation (inventory, reserves, commitments, balances, relationships) is a readable fact, not a statement. The rule: if a system has to *tell* you about its state, that is already a point of distrust.

The rules level: "the laws everything runs by are fixed and visible." Parameters are either built into the design of the system or frozen once at creation and then cannot be changed by anyone. There are no rules that can be "tweaked," so the third level of honesty, the honesty of the process of changing the rules, is reduced to zero: no such process exists. Only monotonic changes that don't touch any rule are allowed (for example, topping up a pre-announced resource "only upward"): a readable value changes, the law does not. This fits the known principles of neutral mechanisms: publicly verifiable execution, simplicity and infrequent change. [7] Buterin counts guaranteed execution according to programmed rules among the properties of Ethereum that hold throughout its evolution from 2015 to 2030. [40]

9 How verifiability is implemented: principles and examples

The principles in this section are not tied to any one technology. But today the most complete environment for them is public blockchains with executable contracts: Ethereum and compatible networks, where both state and rules are expressed as code and public data that any node can reproduce. [10] The examples below come from this environment; their meaning carries over to any system that can show a machine primary data instead of claims.

9.1. State instead of statements

Reserves, balances, commitments and relationships are stored as the contract's public state and read directly. An agent doesn't need to ask the system "how much do you have?"; it reads that for itself. Reports, press releases and numbers on a storefront stop being the source of truth and become just a translation of the primary data. BlackRock describes it this way: tokenized assets are expressed as machine-readable transaction data and governing rules, the network verifies authorization and the validity of the operation, and smart contracts apply conditions specific to the asset. [42]

9.2. Frozen rules and invariants

Parameters are fixed at creation, and after that no call can change them; in a running system there are no addresses with special rights. Formally, the "rules level" is defined by a set of invariants that can be checked by audits and automated tests: total issuance of a resource over any period does not exceed a pre-announced budget; a commitment cannot be withdrawn before its date; no parameter can be changed after creation; no registry entry can be overwritten. An invariant is a rule turned into a verifiable statement about every possible future state.

9.3. Commitment instead of loyalty

One consequence of automated comparison is that switching suppliers becomes cheap (Section 10): loyalty built on habit doesn't hold a machine. A verifiable environment offers a replacement: **a commitment**, a condition the participant accepts voluntarily, with terms and parameters known in advance and recorded in a public registry. An example: locking funds for a period the participant chose. In economic theory, this is a special case of a credible commitment, a "hostage" that makes a promise reliable [23]; in relationship marketing, commitment and trust have long been seen as the key supports of long-term relationships. [16]

The test that separates an honest commitment from a "kill switch" is the question "who decides?" If the exit conditions were chosen by the participant in advance, were known before entry and cannot be changed after the fact by anyone (neither the system nor its creators), the deal is protected in both directions. A registry of such commitments turns a promise into a schedule: not "the resource is here, but it might leave," but "amount X until date Y." For a machine buyer, this is a direct replacement for loyalty: instead of a feeling, there's a verifiable term.

> **Author's note.** We are not claiming that emotional loyalty disappears: a person can still be loyal to a brand and to their own agent (compare the model in which emotional capital and verifiable execution work together [32]). Our claim is narrower: in the part of the choice that a person has delegated to a machine, the role of loyalty passes to verifiable commitments.

9.4. Authentic demand as a verifiable signal

Nearly all the metrics an agent uses to evaluate a project (number of users, trading volume, locked liquidity, activity) are indirect indicators of demand, and all of them can be inflated. The scale of the problem is well documented. An NBER study found that on unregulated crypto exchanges, wash trading made up on average more than 70% of reported volume. [46] Chainalysis estimated suspected wash trading on Ethereum, BNB Smart Chain and Base in 2024 at about $2.57 billion. [47] During the LayerZero token distribution in 2024, about 800,000 of roughly 6 million addresses were preliminarily flagged as farms created for the airdrop, and after the snapshot the number of daily messages on the protocol fell by about 90%. [48] An agent that takes such numbers at face value gets the most important thing wrong: whether the product has real demand. So the claim "we have lots of users" becomes a verifiable fact only when the authenticity can be checked along with the number.

Authenticity shows in the patterns any activity leaves in public history: the age of addresses and how varied their histories are; shared funding sources, that is, clusters of wallets funded from one address; how synchronized and regular the actions are; the variety of counterparties; retention, meaning whether participants come back months later; and the structure of the graph of connections. These are exactly the

methods the industry already uses to filter out address farms before token distributions and to estimate wash trading volume. [47][48] The same asymmetry of costs is at work here: faking plausible organic activity with a long and varied history is expensive, and that cost grows over time.

> **Author's note: organic activity in a machine economy.** When agents themselves become legitimate market participants, "organic" stops meaning "human": agent activity is not fake, yet in blockchain records it's hard to tell apart from scripts. [43] The criterion shifts from "human or bot" to "independent real demand or coordinated imitation." A hundred independent agents paying for a service are real demand; a thousand wallets owned by one party passing funds around in a circle are not. Authenticity can't be strictly proven [9], any heuristic makes mistakes, and analyzing connections runs into privacy (Section 9.8). So the system doesn't issue a verdict itself: it makes the demand data readable, and independent observers with their own methods do the evaluating.

9.5. Simplicity as a requirement of honesty

Elegance and honesty coincide: the simpler the system and the less custom logic sits on top of a proven base primitive, the smaller the surface for hidden trust. Complexity doesn't add value: it adds places where honesty can leak, and more for an observer to check. In security, this principle is known as economy of mechanism. [39] From it follows a second principle: don't reinvent the wheel. Wherever a task has already been solved in a proven way, use the proven solution; every line of custom code adds something an observer will have to check. A telling example from practice is the architecture of Uniswap v2, where the core is deliberately minimal, because contracts with a smaller surface are easier to analyze and can state the desired properties of the system directly. [11]

> **Author's note: which complexity is dangerous.** Buterin distinguishes encapsulated complexity, isolated inside a component, from systemic complexity, which comes from interdependencies between parts of a system [41]; he counts the complexity of zero-knowledge proofs as encapsulated. [40] For machine-verifiable honesty, the danger is above all systemic complexity. Cryptography that is complex but isolated and well proven doesn't contradict this principle.

9.6. Three layers of trust and agent-first interfaces

Core	the subject of machine-verifiable honesty: rules, registries, accounting. Self-sufficient: every operation can be performed by direct calls; zero control "knobs," zero participants with special rights.
Periphery	convenience contracts that bundle several steps into one. It holds neither state nor funds between operations and has no rights in the core. You don't need to trust the periphery, because the core doesn't require it: the same steps can be done by hand (the core/periphery pattern [11]).
Storefront	the human interface, a translator from the language of primary data, but not a source of truth. It is a hidden trust point, and an honest system names it. The principle: the storefront shows only values derived from the core, ideally with a verification path for every number.

The principle running through all interfaces is **agent-first design**: the primary consumer of the system's state is a machine (an autonomous agent, an observer's script, an AI assistant), and a person comes second. In practice, this means aggregating read methods, complete data in events so no one has to reconstruct history piece by piece, and a documented verification path for every value. BlackRock's industry report (2026) describes the same logic: structured, machine-readable representations give agents a more direct interface to blockchain data than fragmented legacy systems, and an agent can evaluate balances and rules before an authorized transaction and then verify settlement. [42] The human storefront is a layer on top of the same machine layer, not a separate channel of truth. In the terms of Section 7, this is Machine Interpretability, Machine Comparability and Agent Conversion all at once, built into how the system works rather than how it is described.

9.7. The cost of verification as a measurable quantity

The asymmetry of costs depends on verification being cheap, so that cheapness needs to be measured. One way to do it is an external verification scenario that an independent observer runs on their own:

1. Read the fixed rules and parameters.
2. Read the current state and compare it with what is claimed.
3. Compute future commitments and resource issuance from the fixed rules.
4. Check the history of permitted changes (for example, top-ups) against current balances.
5. Assess the authenticity of activity from public history using your own method.
6. Check any number on the storefront against the core.

The time, computing resources and expertise needed to complete the scenario are the cost of verification. It can be measured experimentally, including in an adversarial setup where some participants try to cheat the system, with the aim of confirming the asymmetry of costs empirically rather than just asserting it.

> **Author's note: how to compare systems.** For comparison, a rough composite estimate from three quantities can be used: the number of trusted dependencies, that is, trust points without which the scenario can't be completed (storefront, external data sources, intermediaries); the number of requests to primary sources needed to complete the scenario; and the time and computing costs for an independent observer. The first quantity matters more than the others: a system with fewer trust points is preferable even if verification costs more, because expensive verification can be made cheaper with tools, while a trust point can't be removed without changing the system itself.

9.8. Transparency and privacy: verifying without revealing

Full readability of state runs up against participants' right to privacy and against trade secrets: a public history of operations reveals participants' connections and behavior. The conflict doesn't require giving up verifiability. Zero-knowledge proofs make it possible to prove a property ("reserves are at least X," "the participant meets the condition," "all commitments are backed") without revealing the underlying data; proofs of solvency that don't expose customer balances are built on this principle. [37] The W3C Verifiable Credentials 2.0 standard was designed from the start to be both machine-verifiable and privacy-respecting. [33] Buterin (2026) describes a future Ethereum architecture in which zero-knowledge proofs

play a key role at the base layer, and he looks separately at privacy of writes, reads and the network layer. [40] For a buyer agent, privacy of reads matters most: the agent's own verification queries reveal its owner's intentions.

> **Author's note.** Machine-verifiable honesty requires that *claims* be verifiable, not that *all data* be public. The more sensitive the data, the more sense it makes to replace disclosure with a proof of the property.

9.9. Who actually does the checking

The phrase "verification is almost free" holds for *reading* data, but not always for *interpreting* it. Today a language-model agent doesn't check claims against a blockchain by itself: it needs tools such as nodes and RPC access, indexers, analytics services and specialized validators (in ERC-8004 this is a separate validation registry [36]). Each such intermediary becomes a new trust point and a new market participant with its own incentives.

The infrastructure is moving to reduce this dependence. As Buterin (2026) describes it, block verification in Ethereum is shifting from downloading and re-executing everything to checking a SNARK proof and sampling data availability (PeerDAS), and light clients will be able to verify the correctness of computation as well as consensus; for the strongest guarantees you still need your own node, but the requirements for it are becoming much lower. [40] In this sense, "almost free verification" is still a direction the base infrastructure is moving in, rather than a state already reached. The gap between reading and interpreting is already visible in practice: an analysis by TRM Labs found that in blockchain records, AI agent payments are hard to tell apart from ordinary scripts and scheduled jobs. [43]

> **Author's note.** Cheap verification alone isn't enough; machine-verifiable honesty also needs **many independent verifiers**: any observer should be able to reproduce a check, rather than depend on a single service. The design consequence: ship a documented verification path along with the data, so anyone can repeat the check independently.

10 How the object of optimization differs

Dimension	Person as the audience	Machine as the audience
Perception	Images, language, emotions, social signals	Structure, unambiguity, semantics, data availability
Comparison	Often nonlinear, contextual, heuristic	Fast matching of attributes, constraints and evidence
Trust	Brand, reputation, social proof	Data provenance, consistency, policy, verifiability
Persuasion	Argument, image, emotional response	Lowering the cost of verifying a claim
Choice	A mix of rational and emotional	Optimization against given criteria, rules and constraints

Dimension	Person as the audience	Machine as the audience
Path to purchase	Ad → interest → research → decision	Query → filtering → comparison → recommendation → action
Switching suppliers	Can be held back by habit and attachment to the brand	Can be very cheap if the agent easily compares alternatives
Retention	Loyalty, habit, emotional connection	Verifiable commitments with terms and conditions known in advance

> **This does not mean the person drops out of the model.** The person still defines the need, constraints, budget, preferences and how much to delegate. The machine changes mainly the mechanics of search and choice.

The last two rows of the table are linked: the cheaper switching is for a machine, the more valuable whatever replaces loyalty becomes for the seller. In a verifiable environment, that role is played by a commitment (Section 9.3). It doesn't hold on to the customer the way habit does; it puts a guarantee on the table that the agent can factor into its comparison.

11 The limits of machine verification

Machine-verifiable honesty works flawlessly within its domain and runs into trust that can't be removed in several places. An honest system names these places and narrows them, rather than pretending they don't exist.

11.1. The oracle problem: where the system meets reality

Everything that belongs to the outside world (the real value of the settlement asset, the fact of delivery, the quality of goods) can be verified by a machine only up to the system's boundary; beyond it begins trust in an external data source. The same industry practice offers a telling example: customer and agent checks (KYC, KYA) happen off-chain, and only their result goes on-chain, so its reliability depends on whoever did the checking. [42] This can't be eliminated, but it can be narrowed: there should be as few external dependencies as possible, and each should be as simple and transparent as possible.

11.2. The problem of meaning: where the system meets value and the future

The system verifies price, ownership and compliance with the rules, but not value and not intent. Whether anyone really needs a product can't be verified by a machine, in principle.

> **Architecture can guarantee that no one lies about what happened,** but it can't guarantee that what happens is meaningful.

Verifiable authenticity solves the problem of trust, but not the problem of why something exists at all. This is exactly where machine-centric marketing meets human-centric marketing again: meaning, need and value are shaped by people, and this part of marketing is not handed over to the machine, neither as the audience nor as the checker.

11.3. The storefront and trust points outside the core

The interface through which a person sees the system, as well as any state before launch (for example, gathering resources before the system starts running), are named trust points. They can't be removed completely, but they can be narrowed: the storefront has to derive every number from the core, and preliminary stages have to end with a single verifiable operation, after which the trust point disappears.

11.4. Scope of application

The strong form of machine-verifiable honesty works fully where state and rules exist inside a verifiable environment: digital and financial assets, contracts, registries. For physical goods and services, almost any significant claim runs into the oracle boundary (Section 11.1): trust doesn't disappear but shifts to whoever issues the attestation (the manufacturer, a lab, a certification body). Verifiable credentials make that trust traceable and revocable, but they don't remove it. [33] So for most traditional commerce we are talking about the *weak* form of the concept: verifiable origin of claims, not verifiability of the facts themselves.

11.5. Less obvious trust points

Beyond external data and the storefront, any verifiable system keeps less visible trust points: the compiler and whether the published code matches the deployed code; network infrastructure (for example, the sequencer in layer-2 networks); the data access provider through which the agent reads the state; and finally the AI agent itself as the interpreter of the data. Each can be narrowed through source code verification, running your own node or using several independent sources, but an honest system still names them.

11.6. The substitution risk: verifiability as a metric

As soon as the cost of verification becomes a metric, there's a temptation to optimize the appearance of verifiability instead of verifiability itself. That's the classic Goodhart effect. GEO research shows that machine readers already reward the signs of evidence. [15] So the distinction that matters is between verifiability of *any* claim and verifiability of *significant* claims: it's easy to make something verifiable that doesn't matter for the choice. This is the operational side of the problem of meaning (Section 11.2).

11.7. The price of immutability

The immutability of rules that the rules level depends on (Section 8.5) comes at a price. Frozen code can't be fixed if a vulnerability is found in it, and the history of decentralized finance shows this is a real risk. In practice, the ban on changing rules is often worked around through migration: a new contract is created and participants are asked to move to it, so the rules change after all, just by a back door. An honest system names this trade-off and narrows it: with a minimal core (Section 9.5), with audits and formal verification of invariants (Section 9.2), and, if an upgrade mechanism is truly needed, by describing it explicitly in the rules themselves, with conditions known in advance and a delay long enough for those who disagree to leave. In that case, the honesty of the process of changing the rules is no longer reduced to zero and becomes something to verify in its own right.

> **The principle of limits:** claims about honesty are acceptable only together with explicit limits on where they apply. A named limit is part of verifiability; a hidden limit is a violation of it.

12 Ethics: marketing to machines and manipulating machines

Machine-centric marketing has a dark twin. If a buyer agent is sensitive to position, wording and embedded instructions (Section 2.3), then "optimizing for the agent" easily turns into exploiting its weaknesses: tailoring descriptions to the model's biases, fake trust signals, prompt injection in a product listing. Experiments show that such tricks already work. [26][27] The one who gets hurt is the person who trusted the agent.

> **Author's note: where the line is.** Machine-centric marketing, as this paper understands it, is bound by one rule: **you may influence an agent's choice only with what the agent can verify**. Making verification cheaper, a clear data structure and verifiable commitments are legitimate means. Exploiting the model's biases, unverifiable claims dressed up as verified ones, and embedded instructions are manipulation, however well they work. In this logic, machine-verifiable honesty protects the choice a person has delegated, and it is a marketing advantage too.

This also creates a duty in the other direction: since agents act on behalf of people, the environment itself should make it easier for them to resist manipulation, by giving them primary data instead of descriptions and a verification path instead of assurances.

13 The concept in summary

The move from digital commerce to agentic commerce creates a new object for marketing: **the machine intermediary**, which does more than deliver a message to the consumer: it takes part on its own in discovering, comparing and choosing an offer.

So on top of traditional human-centric marketing, a new task appears, machine orientation: making the value of a product **visible, interpretable, comparable, verifiable and preferable** to an AI agent.

It makes sense to think of this set of tasks as **machine-centric marketing** (Machine-Centric Marketing).

> **In this sense the term doesn't compete with Agentic Commerce, Machine Customer or Machine-Readable Commerce; it connects them at the level of marketing theory:**
>
> Machine Customer answers the question "who chooses?";
>
> Agentic Commerce answers "how does trade happen?";
>
> Machine-Readable Commerce answers "what should the information environment be like?";
>
> Machine-Centric Marketing answers "how should marketing work so that this machine buyer chooses our offer?"

The answer to the last question comes down to trust, and **a machine's trust can't be instilled: it can only be made unnecessary by replacing it with verification.**

Hence the paper's second concept, **machine-verifiable honesty**: an environment where verification is almost free, deception is impossible or immediately visible, and honesty becomes the path of least resistance. It answers the question "why can a machine trust this offer?" and provides a measurable quantity: **the cost of verification**.

From this follows a conclusion shared by both parts: for a machine audience, **simplicity is not an aesthetic choice but a requirement of honesty**. Every added complication adds surface where honesty can leak, and more for an observer to check.

The most convincing offer for a machine is the one that is most verifiable in what matters: a minimum of custom logic on top of a proven foundation, nothing that can be "tweaked," and explicitly named limits where machine verification gives way to trust that can't be removed (external data and meaning).

> **Author's note: why bring the fields together.** Marketing sees the machine buyer as a new audience, economics sees an old problem of information asymmetry in a new form, AI researchers see a vulnerable executor, and Ethereum engineers see a trust minimization problem. All of these describe one process. As long as they stay apart, the marketer doesn't know that economists have formalized "trust," the economist doesn't know that verifiable disclosure can already be built, and the engineer doesn't know they are building marketing infrastructure. We see the value of this paper in showing one picture from different sides and giving it a name.

14 Sources and terminology check

> **Limits of the check:** terminology in this area is changing fast. The conclusion that Machine-Centric Marketing / Machine-Oriented Marketing and Machine-Verifiable Honesty are not yet generally accepted standard names is based on a web search of English-language sources (without full-text academic databases); it is not a claim that these expressions are never used anywhere.

> **The concept in one formula:**
>
> *Human-Centric Marketing → Machine-Centric Marketing; Agentic Commerce: the environment; Machine Customer: the audience; Machine-Readable Commerce: the infrastructure; Machine-Verifiable Honesty: the foundation of trust.*

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This paper was prepared as part of the Romul Protocol project; the principles it sets out are used in the project's development.

About authorship. The name "Human and Machine" literally describes how this paper was written. The ideas, the concept and the hands-on experience belong to the human; finding and checking sources, comparing with the literature, structure and editing were done together with a language model. Decisions about content and responsibility for the text remain with the human.

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